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CMPT 351 Assignment 2 Marking Guide
Assignment must be done individually or in pairs.

Total: [110] Marks

[100] Light Sampler
   [10] Design/Style
      Good modular design such as:
      - clean interfaces (.h clear enough to not need to read .c files)
      - hiding all threads inside modules
        (or potentially a module of threads, if all info about threads hidden)
      - modules and exported functions have great names
      Code Quality
      (Up to a -25 mark penalty for poor code quality. Some specifics:
      - Each .h file must have a description about what that module does. (Max -5)
      - Functions which are not in the .h file have internal linkage. (Max -5)
      - Great function, and variable names. (Max -5)
      - No magic numbers; 0, 1, -1 2 often OK. (Max -5)
      - No very large cryptic functions, no deep nesting of code, great indentation, .... (Max -5)
   [30] Sampling
        Correct data sampling: correct values read and stored, reasonable timing.
        Must achieve at least 250 samples per second
        Test by using the UDP "history" command, viewed using the python script.
        Test by blocking/unblocking light to ensure dips stops/start, etc.
   [10] Dip Detection
        Correctly uses thresholds and hysteresis to detect dips reliably up to ~20hz
        Check the value as it is displayed on serial port and/or terminal.
        Can test with a phone strobe light app, or using LED
   [10] Terminal
       Correct output on the terminal; including:
       - Samples/sec, Flash Hz, and dip count
       - timing for sampling (min, max, avg, # samples)
       - 10 samples collected during the previous second, ~evenly spaced across data
   [15] Reliable rotary encoder operation:
           - [2] Start at 10Hz
           - [5] Correctly counts 24 changes on full rotation
           - [5] Correctly counts when spun each direction quickly
           - [3] Cannot go below 0Hz, When at 0Hz ticking clock-wise increases to 1Hz
   [10] Flash LED emitter at correct frequency
           - Must flash down to 1Hz
   [15] UDP socket
        [12] 2 marks per command: help/?, count, length, dips, <enter>, stop
        [ 3] Correctly displays error message for bad commands, such as "oops"
   [ 0] Memory
        Up to a -20 penalty applied for memory errors found with valgrind.
        Note: No deduction for memory still reachable for call stacks involving pthread_cancel_init
or gpio library.
        Note: No other marking to be done with valgrind on because it significantly affects
application timing.

[10] Debug noworky
    [5] GDB text capture
       - Show reasonable detail to discover the bug.
       - Expecting use of at least: bt, list, up, print.
    [3] Graphical Debugger
       - Screenshot shows noworky being debugged through Eclipse (or another graphical debugger).
    [2] noworky.c
       - Must have corrected and commented the change.
       - If the bug is resolved in a different (yet equally direct way) as
         expected, it will also be worth full marks.

       (Graphical debugger screenshot will not show the sequence of steps used to debug
         the application; just that the debugger is connected)
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Forward to instructor if...
- Material is suspiciously similar to another submission.
- Inconsistent submission (inconsistent trace indicating results are forged).
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